Theory of Computation

Classifications of Algorithms in Research

Lecture ~~31_32~~ 33

Gaurav Raj

Polynomial

Real Knapsack

B-S

5	6	8	12
25	50	70	50

2^n 20

Exponential T.C

0/1 Knapsack $\rightarrow 2^n$

Travelling Salesman $\rightarrow 2^n$

Hamiltonian Cycle $\rightarrow 2^n$

Graph Coloring $\rightarrow 2^n$

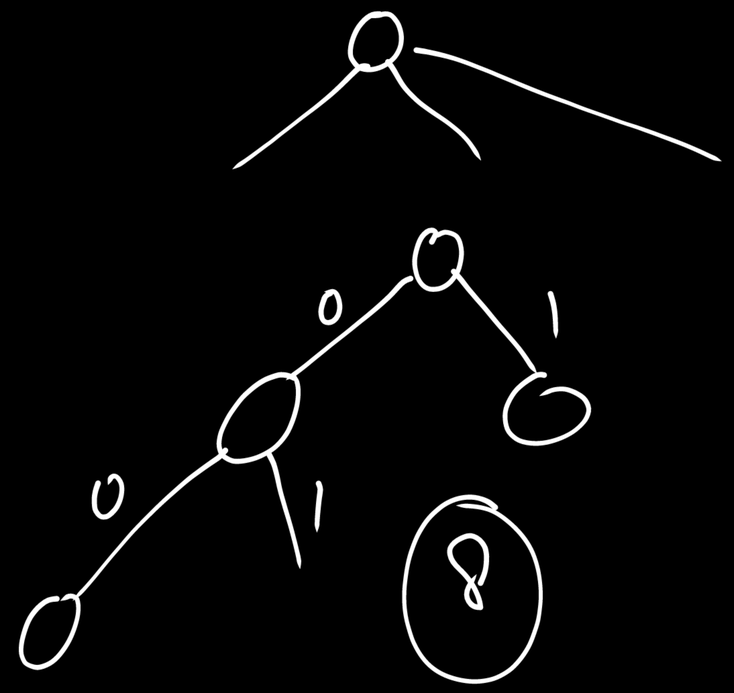
DTM

$$(a \wedge b \wedge c) \wedge (\bar{a} \wedge \bar{b} \vee c) = T$$

CNF →

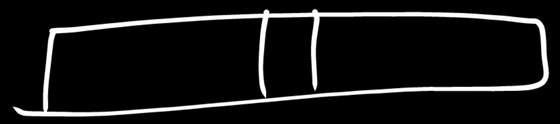
a	b	c
0	0	0
0	0	1
1	1	1

$2^3 = 8$



free

$a = 5$



```
NewSearch() {
    j = ndf();          → O(1)
    if (j == a)         → 1
    {
        success()       — 1
    }
    else
        fail()          — 1
}
```

magical steps

$O(1)$

NP Sol^n

P Solution $O(n)$

Cook

CNF-Sat → 2^n

⇓

NP-Hard

Convert Polynomial time

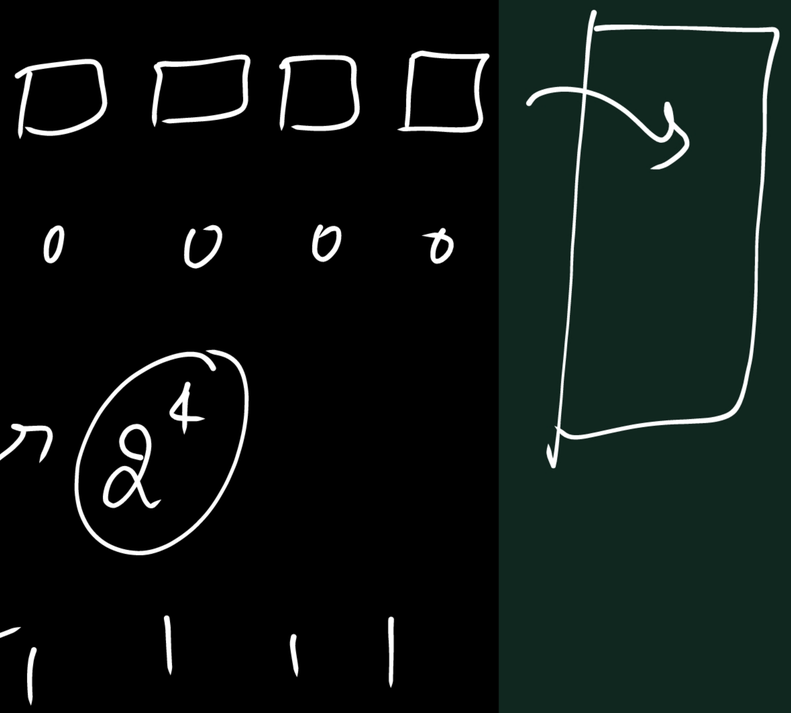
NP-H

T.S (2^n)

Convert P. time

Research

0/1 Knapsack

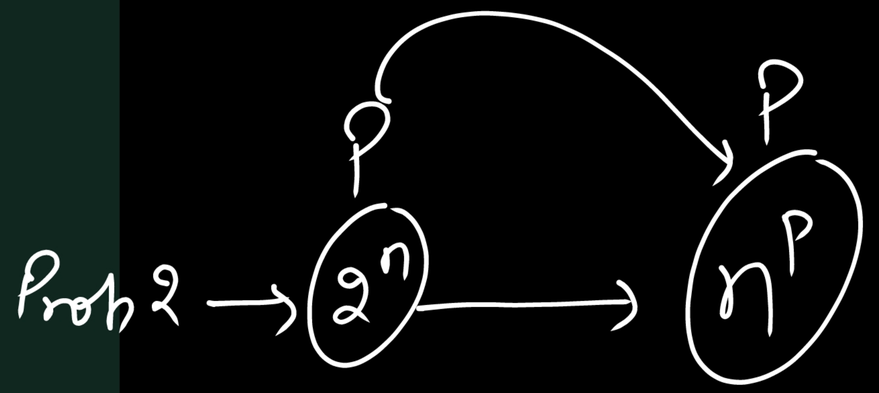


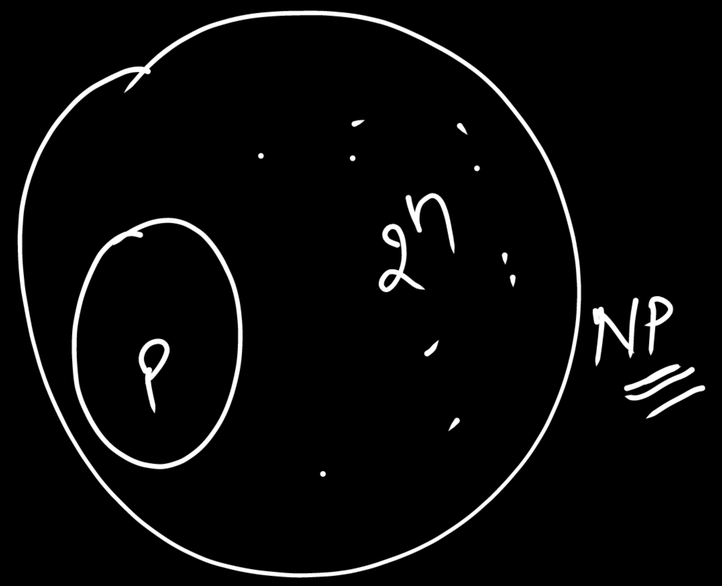
2^4

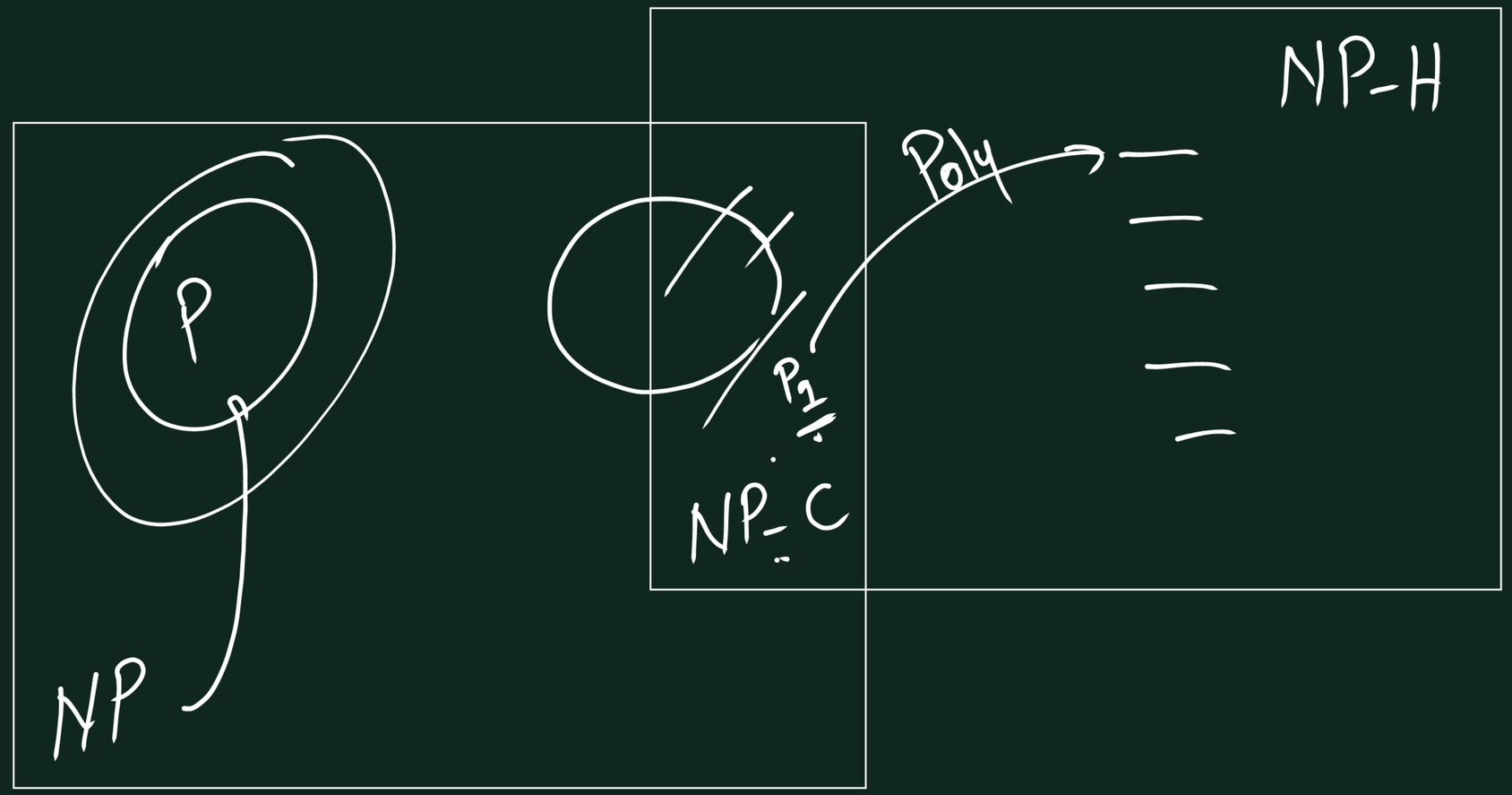
NP-C

(i) NP-H

(ii) NP-Solution






NP-H
Poly
P
NP
NP-C

Thank You